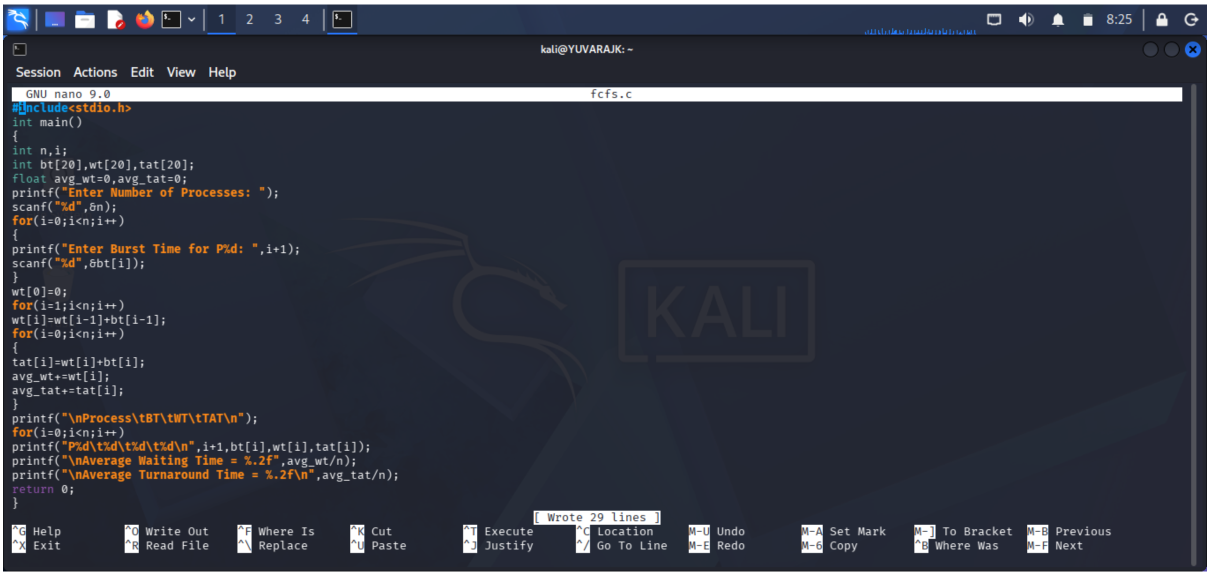
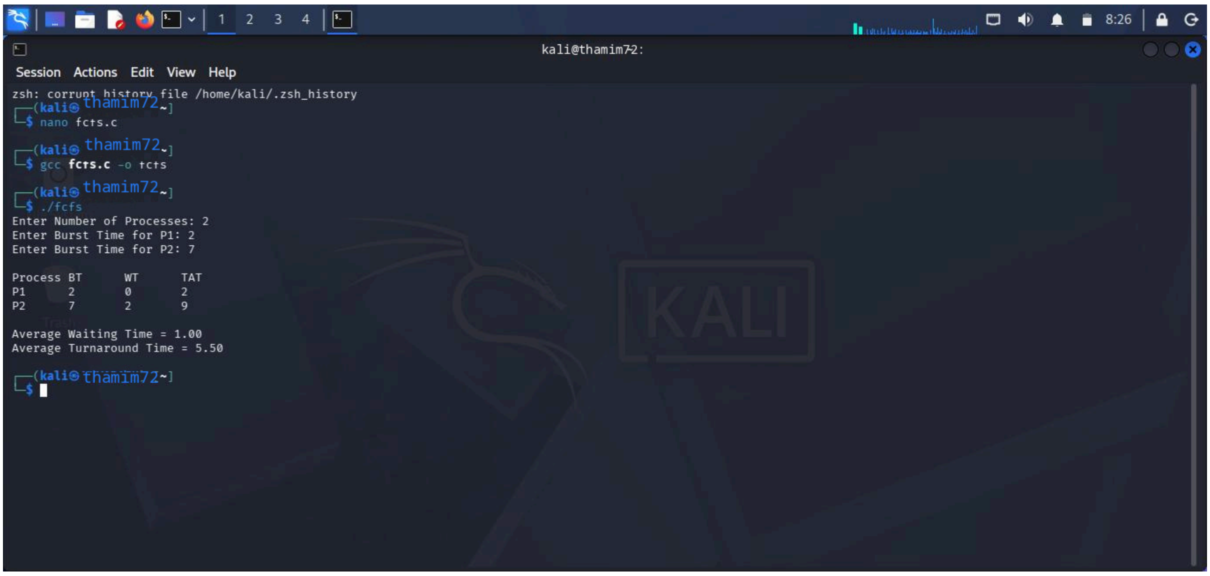
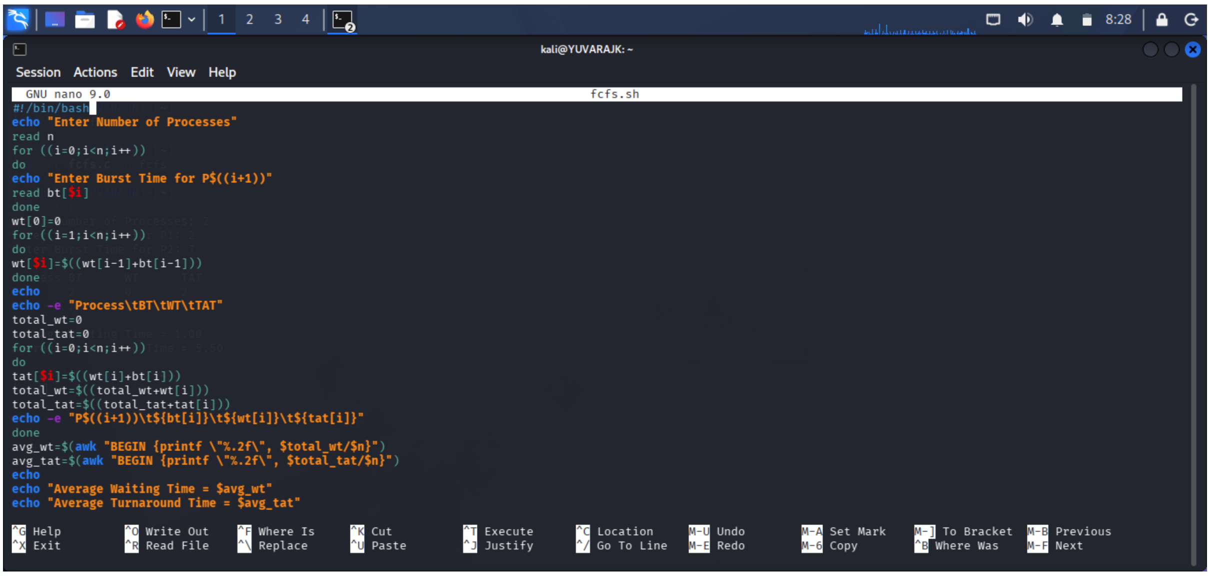
C PRROGRAMMINNG : FCFS



OUTPUT :

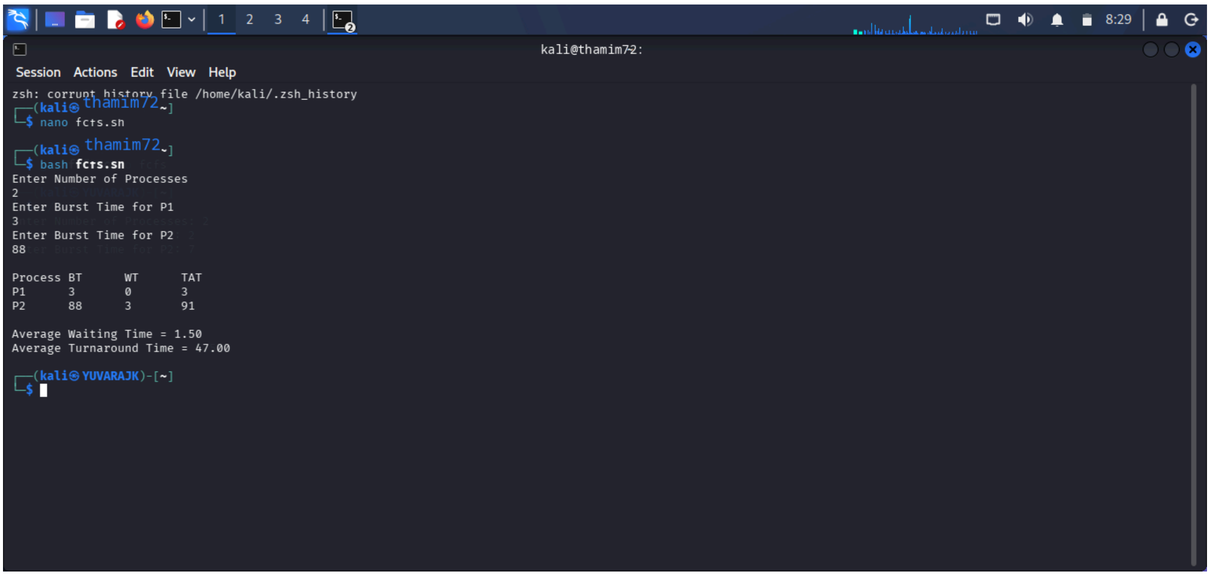


SHELL PROGRAMMING :FCFS



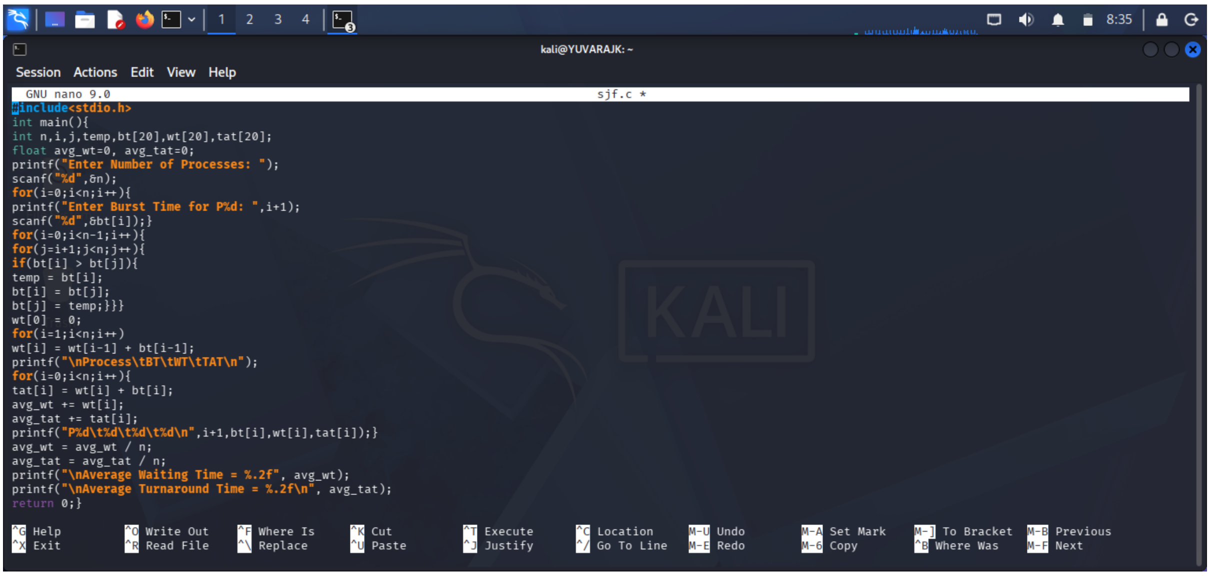
```
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
done
wt[0]=0
for ((i=1;i<n;i++))
do
wt[$i]=$((wt[i-1]+bt[i-1]))
done
echo
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i=0;i<n;i++))
do
tat[$i]=$((wt[i]+bt[i]))
total_wt=$((total_wt+wt[i]))
total_tat=$((total_tat+tat[i]))
echo -e "P$((i+1))\t${bt[i]}\t${wt[i]}\t${tat[i]}"
done
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt/$n}")
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat/$n}")
echo
echo "Average Waiting Time = $avg_wt"
echo "Average Turnaround Time = $avg_tat"
```

OUTPUT :

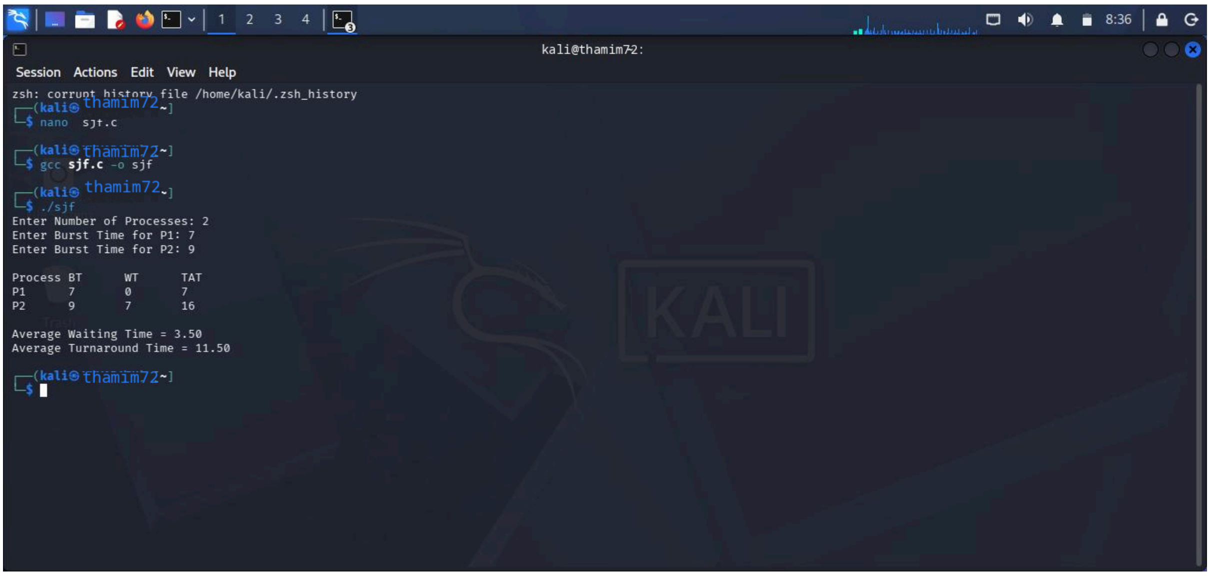


```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿ thamim72 ~]
└─$ nano fcfs.sh

┌──(kali㉿ thamim72 ~]
└─$ bash fcfs.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
88

Process BT      WT      TAT
P1      3       0       3
P2      88      3       91

Average Waiting Time = 1.50
Average Turnaround Time = 47.00

┌──(kali㉿ YUVARAJK)-[~]
└─$
```

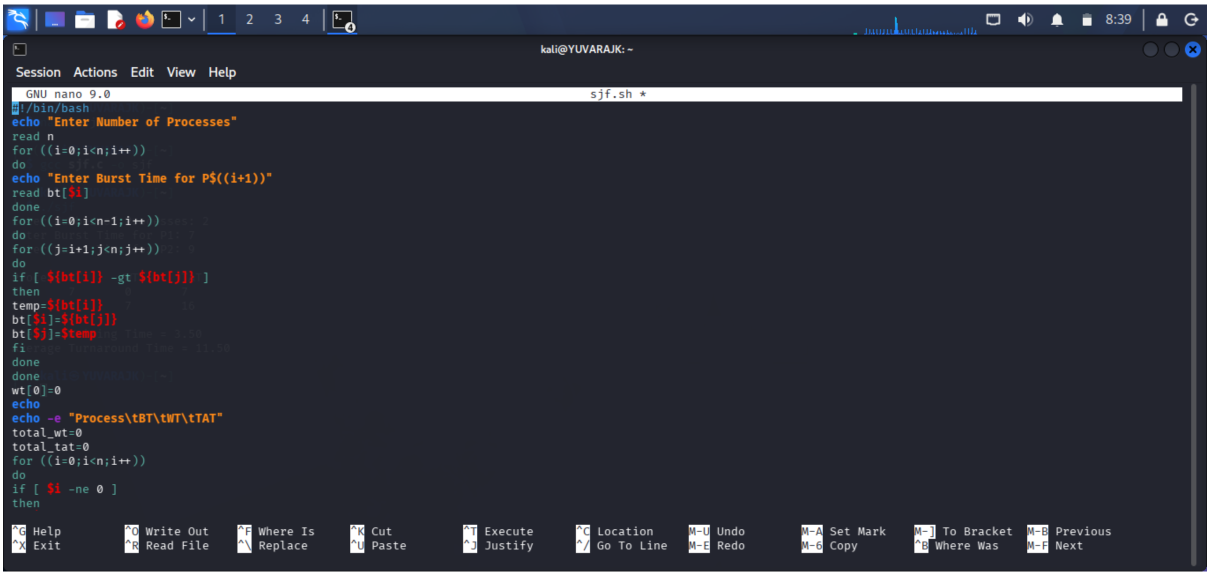
C PROGRAMMING :SJF



OUTPUT :

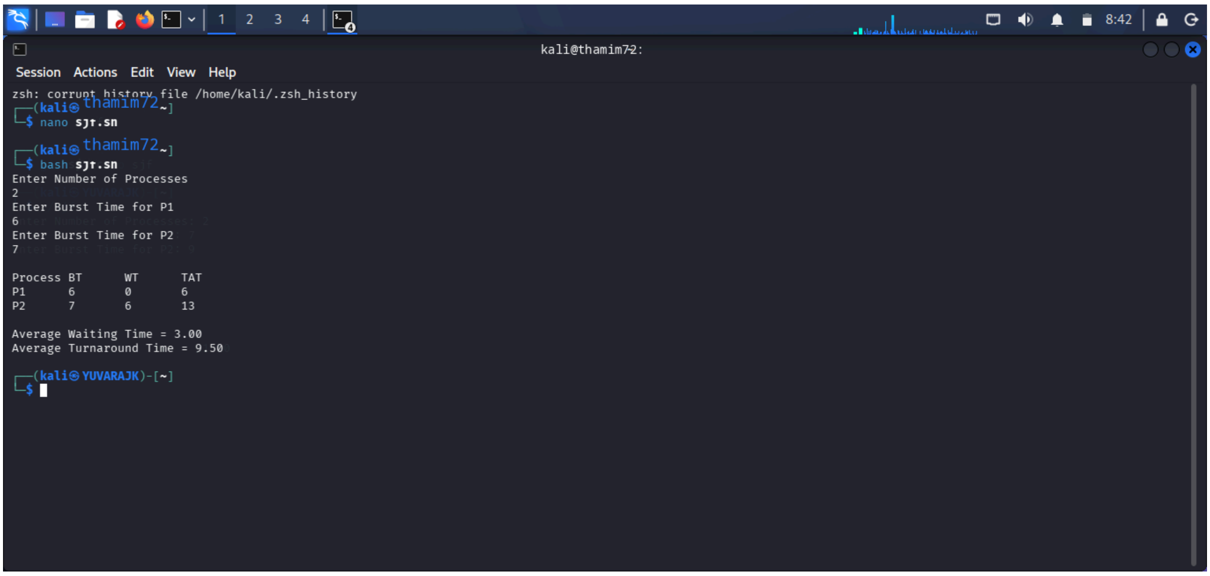


SHELL PROGRAMMING : SJF



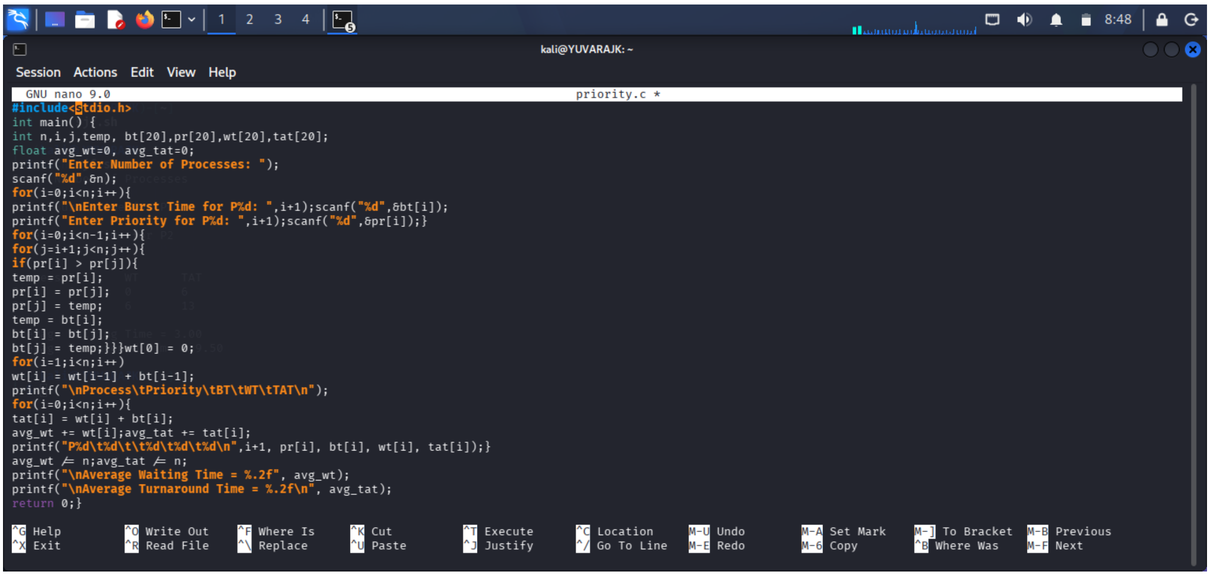
```
GNU nano 9.0                                sjf.sh *
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
done
for ((i=0;i<n-1;i++))
do
for ((j=i+1;j<n;j++))
do
if [ ${bt[i]} -gt ${bt[j]} ]
then
temp=${bt[i]}
bt[$i]=${bt[j]}
bt[$j]=$temp
fi
done
done
wt[0]=0
echo
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i=0;i<n;i++))
do
if [ $i -ne 0 ]
then

^G Help      ^O Write Out  ^F Where Is  ^K Cut    ^T Execute  ^C Location    M-U Undo  M-A Set Mark  M-] To Bracket  M-B Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy      ^B Where Was    M-F Next
```

OUTPUT :



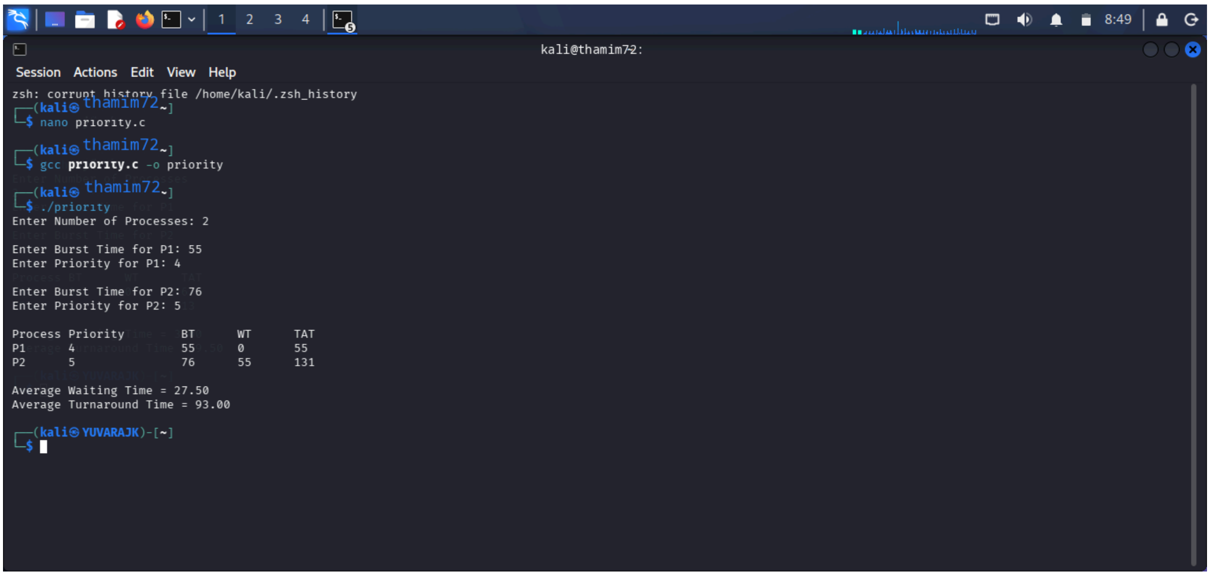
```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿thamim72)-[~]
└─$ nano sjf.sh

┌──(kali㉿thamim72)-[~]
└─$ bash sjf.sh
Enter Number of Processes
2
Enter Burst Time for P1
6
Enter Burst Time for P2
7

Process BT      WT      TAT
P1      6       0       6
P2      7       6       13

Average Waiting Time = 3.00
Average Turnaround Time = 9.50

┌──(kali㉿YUVARAJK)-[~]
└─$
```

C PROGRAMMING : PRIORITY SCHEDULING



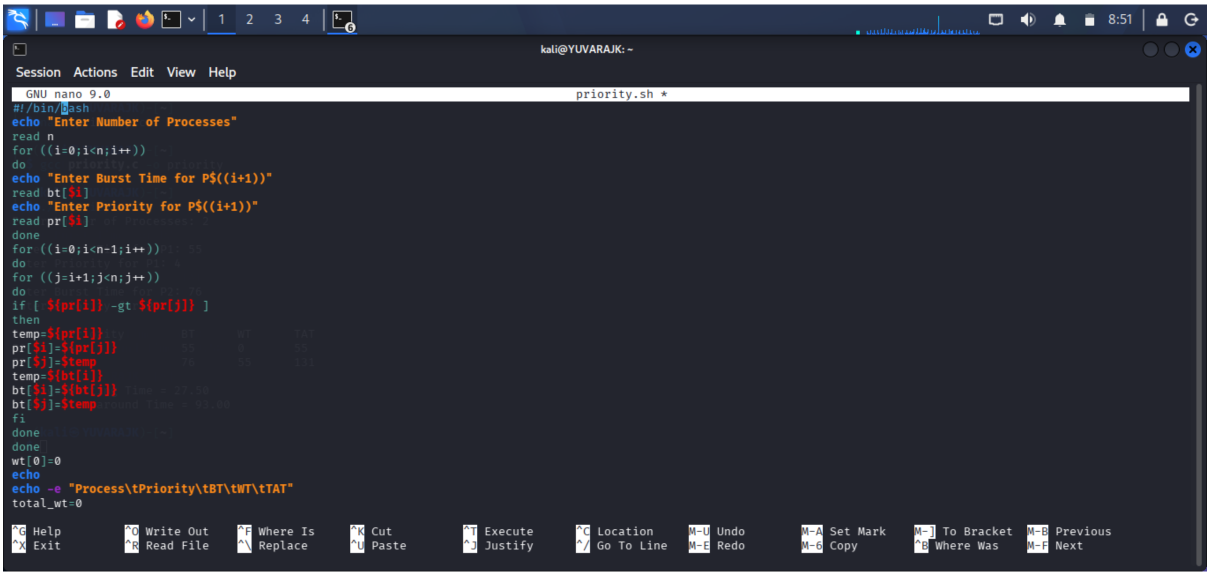
```c
#include<stdio.h>
int main() {
int n,i,j,temp, bt[20],pr[20],wt[20],tat[20];
float avg_wt=0, avg_tat=0;
printf("Enter Number of Processes: ");
scanf("%d",&n);
for(i=0;i<n;i++){
printf("\nEnter Burst Time for P%d: ",i+1);scanf("%d",&bt[i]);
printf("Enter Priority for P%d: ",i+1);scanf("%d",&pr[i]);}
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(pr[i] > pr[j]){
temp = pr[i];
pr[i] = pr[j];
pr[j] = temp;
temp = bt[i];
bt[i] = bt[j];
bt[j] = temp;}}}wt[0] = 0;
for(i=1;i<n;i++)
wt[i] = wt[i-1] + bt[i-1];
printf("\nProcess\tPriority\tBT\tWT\tTAT\n");
for(i=0;i<n;i++){
tat[i] = wt[i] + bt[i];
avg_wt += wt[i];avg_tat += tat[i];
printf("P%d\t%d\t\t%d\t%d\t%d\n",i+1, pr[i], bt[i], wt[i], tat[i]);}
avg_wt /= n;avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;}
```

OUTPUT :



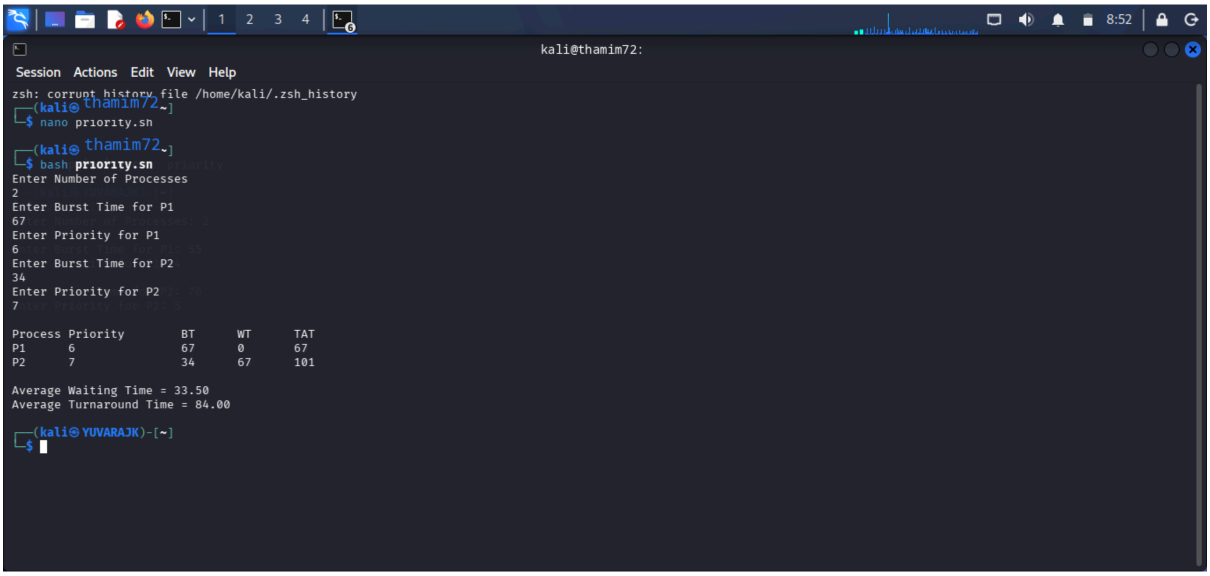
```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿thamim72)-[~]
└─$ nano priority.c

┌──(kali㉿thamim72)-[~]
└─$ gcc priority.c -o priority

┌──(kali㉿thamim72)-[~]
└─$ ./priority
Enter Number of Processes: 2

Enter Burst Time for P1: 55
Enter Priority for P1: 4

Enter Burst Time for P2: 76
Enter Priority for P2: 5

Process Priority        BT      WT      TAT
P1      4               55      0       55
P2      5               76      55      131

Average Waiting Time = 27.50
Average Turnaround Time = 93.00

┌──(kali㉿YUVARAJK)-[~]
└─$
```

SHELL PROGRAMMING : SJF



```
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
echo "Enter Priority for P$((i+1))"
read pr[$i]
done
for ((i=0;i<n-1;i++))
do
for ((j=i+1;j<n;j++))
do
if [ ${pr[i]} -gt ${pr[j]} ]
then
temp=${pr[i]}
pr[$i]=${pr[j]}
pr[$j]=$temp
temp=${bt[i]}
bt[$i]=${bt[j]}
bt[$j]=$temp
fi
done
done
wt[0]=0
echo
echo -e "Process\tPriority\tBT\tWT\tTAT"
total_wt=0
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
$ nano priority.sh
$ bash priority.sh
Enter Number of Processes
2
Enter Burst Time for P1
67
Enter Priority for P1
6
Enter Burst Time for P2
34
Enter Priority for P2
7

Process Priority        BT      WT      TAT
P1      6               67      0       67
P2      7               34      67      101

Average Waiting Time = 33.50
Average Turnaround Time = 84.00
```

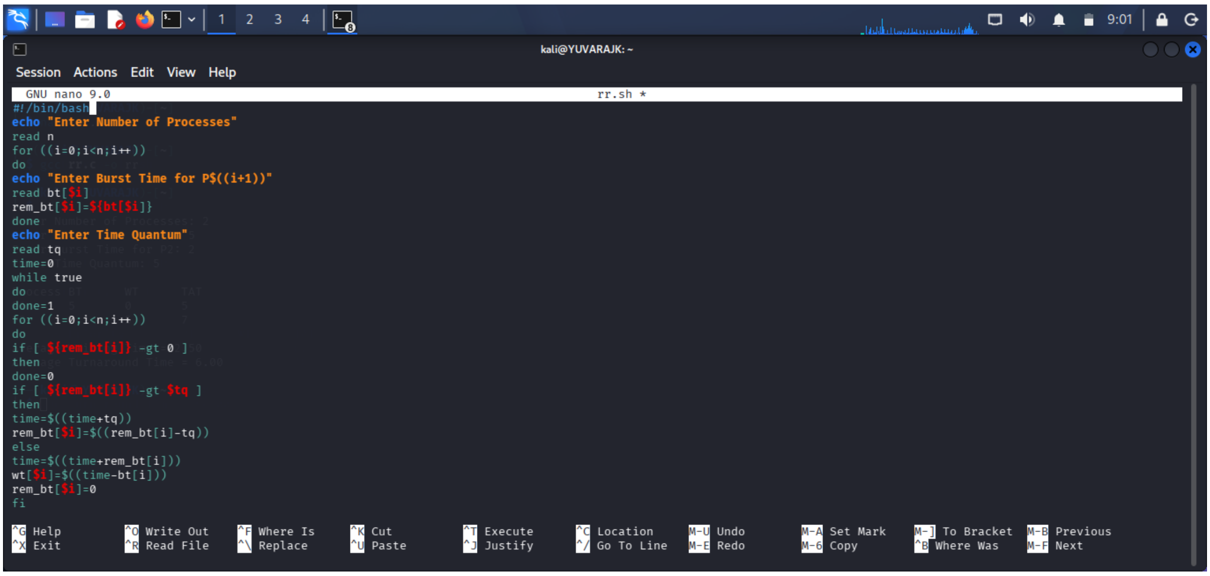
C PROGRAMMING : ROUND ROBIN

```
GNU nano 9.0                         rr.c *
#include<stdio.h>
int main(){
int n, tq, i, bt[20], rem_bt[20], wt[20] = {0}, tat[20],time = 0, done;
float avg_wt = 0, avg_tat = 0;
printf("Enter Number of Processes: ");scanf("%d", &n);
for(i = 0; i < n; i++){
printf("Enter Burst Time for P%d: ", i + 1);scanf("%d", &bt[i]);
rem_bt[i] = bt[i];}
printf("Enter Time Quantum: ");scanf("%d", &tq);
do{
done = 1;
for(i = 0; i < n; i++){
if(rem_bt[i] > 0){
done = 0;
if(rem_bt[i] > tq){time += tq;rem_bt[i]-= tq;}
else{
time += rem_bt[i];
wt[i] = time- bt[i];
rem_bt[i] = 0;}}}
} while(!done);
printf("\nProcess\tBT\tWT\tTAT\n");
for(i = 0; i < n; i++){
tat[i] = bt[i] + wt[i];
avg_wt += wt[i];avg_tat += tat[i];
printf("P%d\t%d\t%d\t%d\n",i + 1, bt[i], wt[i], tat[i]);}
avg_wt /= n;avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;}
```

OUTPUT :

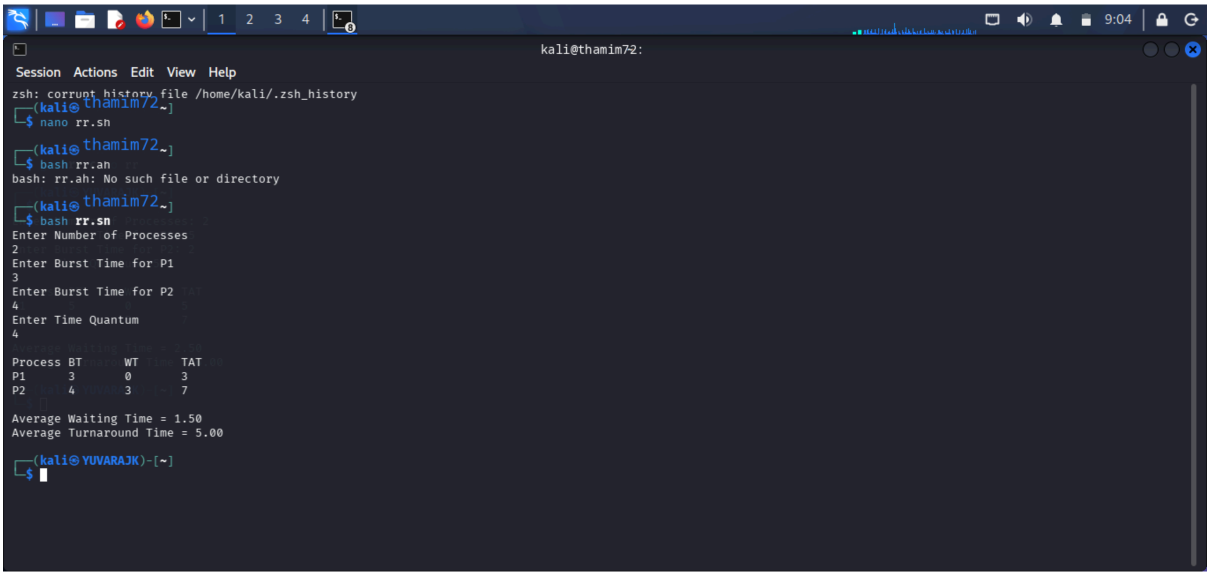
```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿thamim72)-[~]
└─$ nano rr.c

┌──(kali㉿thamim72)-[~]
└─$ gcc rr.c -o rr

┌──(kali㉿thamim72)-[~]
└─$ ./rr
Enter Number of Processes: 2
Enter Burst Time for P1: 5
Enter Burst Time for P2: 2
Enter Time Quantum: 5

Process BT      WT      TAT
P1      5       0       5
P2      2       5       7

Average Waiting Time = 2.50
Average Turnaround Time = 6.00

┌──(kali㉿YUVARAJK)-[~]
└─$
```

SHELL PROGRAMMING : ROUND ROBIN



```
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
rem_bt[$i]=${bt[$i]}
done
echo "Enter Time Quantum"
read tq
time=0
while true
do
done=1
for ((i=0;i<n;i++))
do
if [ ${rem_bt[i]} -gt 0 ]
then
done=0
if [ ${rem_bt[i]} -gt $tq ]
then
time=$((time+tq))
rem_bt[$i]=$((rem_bt[i]-tq))
else
time=$((time+rem_bt[i]))
wt[$i]=$((time-bt[i]))
rem_bt[$i]=0
fi
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿thamim72)-[~]
└─$ nano rr.sn

┌──(kali㉿thamim72)-[~]
└─$ bash rr.an
bash: rr.ah: No such file or directory

┌──(kali㉿thamim72)-[~]
└─$ bash rr.sn
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
4
Enter Time Quantum
4

Process BT      WT      TAT
P1      3       0       3
P2      4       3       7

Average Waiting Time = 1.50
Average Turnaround Time = 5.00

┌──(kali㉿YUVARAJK)-[~]
└─$
```